

Introduction to Knot Theory

(5) Jones Polynomial and Khovanov Homology

Seoul National University

April 24, 2026

Polynomial Invariants: Historical Overview

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 - From Wirtinger presentation (Lecture 2)
 - Seifert matrix $\rightarrow$ determinant $\rightarrow$ Alexander polynomial (Lecture 4)

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 - Discovered new polynomial via operator algebras
- **1987 — L. Kauffman**
 - Reformulated using bracket polynomial

Kauffman Bracket Polynomial

The **Kauffman bracket polynomial** assigns to a link diagram L a Laurent polynomial

$$\langle L \rangle \in \mathbb{Z}[A, A^{-1}].$$

Three defining rules

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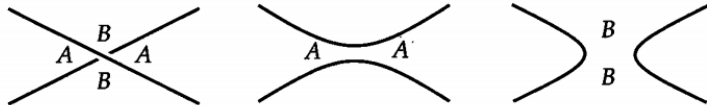
③ **Crossing-expansion rule**

$$\langle L \rangle = A \langle L_A \rangle + A^{-1} \langle L_B \rangle$$

where L_A and L_B are the two diagrams obtained by smoothing the chosen crossing.

Smoothing at a Crossing

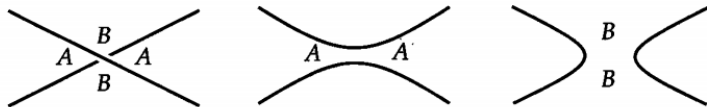
At each crossing, there are exactly **two possible smoothings**.



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Key Observation

Each crossing is replaced by a choice between two local resolutions.

- These choices are the building blocks of the bracket polynomial
- Repeating this at every crossing generates all possible states

How the Computation Works

Step-by-step procedure

- ① Choose a crossing
- ② Replace it by either A-smoothing or B-smoothing
- ③ Repeat this process for all crossings

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Evaluation

- Each circle contributes a factor $(-A^2 - A^{-2})$
- Multiply all contributions and sum over all states

Invariance under Reidemeister II

Idea

We impose that the bracket polynomial is invariant under Reidemeister II.

$$\begin{aligned}\langle \text{X} \rangle &= A \langle \text{Y} \rangle + B \langle \text{Z} \rangle \\ &= A(A \langle \text{X} \rangle + B \langle \text{Y} \rangle) + B(A \langle \text{Z} \rangle + B \langle \text{X} \rangle) \\ &= A(A \langle \text{X} \rangle + BC \langle \text{X} \rangle) + B(A \langle \text{Z} \rangle + B \langle \text{X} \rangle) \\ &= (A^2 + ABC + B^2) \langle \text{X} \rangle + BA \langle \text{Z} \rangle \stackrel{?}{=} \langle \text{X} \rangle\end{aligned}$$

- From the expansion, we obtain the conditions:

$$BA = 1, \quad A^2 + ABC + B^2 = 0$$

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- From the expansion, we obtain the conditions:

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- Solving these gives:

$$B = A^{-1}, \quad C = -A^2 - A^{-2}$$

Invariance under Reidemeister III

$$\begin{aligned}
 \langle \text{Type III move} \rangle &= A \langle \text{Type II move} \rangle + A^{-1} \langle \text{Type II move} \rangle \quad (\text{Now, apply the fact that Type II moves don't change the bracket polynomial}) \\
 &= A \langle \text{Type II move} \rangle + A^{-1} \langle \text{Type II move} \rangle = \langle \text{Type III move} \rangle
 \end{aligned}$$

- Reidemeister III changes only the **order of crossings**, not their number

Invariance under Reidemeister III

$$\begin{aligned}
 \langle \text{Diagram 1} \rangle &= A \langle \text{Diagram 2} \rangle + A^{-1} \langle \text{Diagram 3} \rangle && \text{(Now, apply the fact that Type II moves don't change the bracket polynomial)} \\
 &= A \langle \text{Diagram 2} \rangle + A^{-1} \langle \text{Diagram 3} \rangle = \langle \text{Diagram 1} \rangle
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- Reidemeister III changes only the **order of crossings**, not their number
- Apply the smoothing rule to both sides
- Each side expands into a sum over the same set of states

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the coefficients of all corresponding terms agree

Invariance under Reidemeister III

$$\begin{aligned}
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the coefficients of all corresponding terms agree

Conclusion

The bracket polynomial is invariant under Reidemeister III.

Failure of Reidemeister I

$$\begin{aligned}
 \langle \overline{\text{positive crossing}} \rangle &= A \langle \overline{\text{negative crossing}} \rangle + A^{-1} \langle \overline{\text{positive crossing}} \rangle \\
 &= A(-A^2 - A^{-2}) \langle \overline{\text{strand}} \rangle + A^{-1} \langle \overline{\text{strand}} \rangle \\
 &= -A^3 \langle \overline{\text{strand}} \rangle
 \end{aligned}$$

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 \langle \overline{\text{negative crossing}} \rangle &= A \langle \overline{\text{positive crossing}} \rangle + A^{-1} \langle \overline{\text{negative crossing}} \rangle \\
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- Applying the smoothing rule to a Reidemeister I move gives:

$$\langle L' \rangle = \begin{cases} (-A^3) \langle L \rangle & \text{(positive crossing)} \\ (-A^{-3}) \langle L \rangle & \text{(negative crossing)} \end{cases}$$

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Conclusion

The bracket polynomial is not invariant under Reidemeister I.

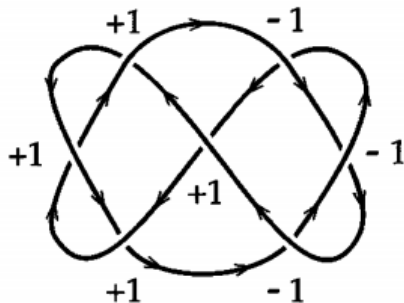
Writhe

Definition

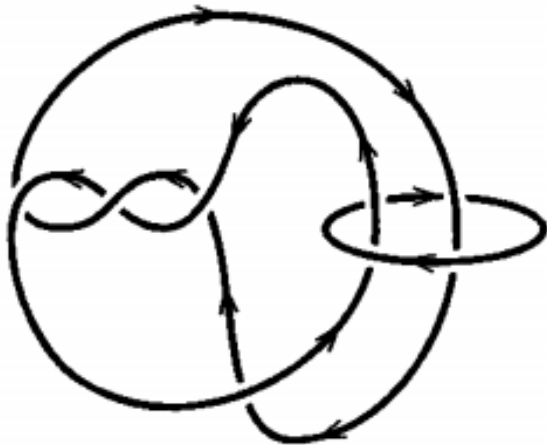
$$w(L) = \sum \text{sign of crossings}$$

- Each crossing contributes $+1$ or -1
- The sign is defined using an orientation
- Reversing all orientations does not change $w(L)$
- Mirror image flips the sign:

$$w(L^*) = -w(L)$$



Write Example



Jones Polynomial

Idea

The bracket polynomial fails under Reidemeister I by a factor of $(-A^{\pm 3})$. We correct this using the writhe.

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$$X(L) = (-A^3)^{-w(L)} \langle L \rangle$$

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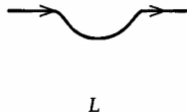
With the substitution

$$A = t^{-1/4},$$

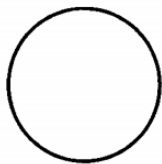
we obtain $V_L(t)$.

Invariance under Reidemeister I

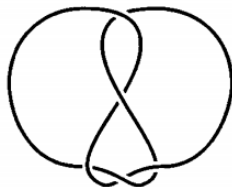
$$\begin{aligned}X(L') &= (-A^3)^{-w(L')} \langle L' \rangle \\&= (-A^3)^{-(w(L)+1)} \langle L' \rangle \\&= (-A^3)^{-(w(L)+1)} ((-A)^3 \langle L \rangle) \\&= (-A^3)^{-w(L)} \langle L \rangle \\&= X(L)\end{aligned}$$



Jones Polynomial Examples (Trefoil, Hopf Link)



$$V(t) = 1$$



$$V(t) = -t^{-1} + 3t^{-2} - 2t^{-3} + 3t^{-4} - t^{-5} - t^{-6}$$

Figure 6.1 Polynomials of the unknot and the 5_2 knot.

State Sum Interpretation

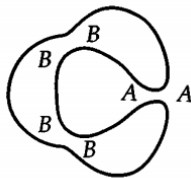
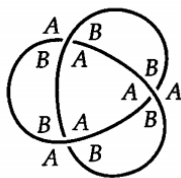
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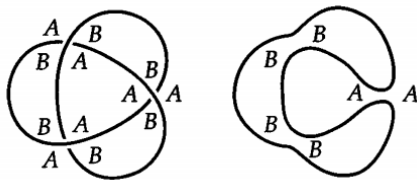
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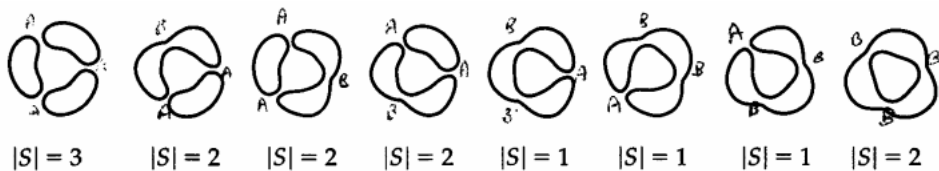
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- Each state contributes

$$A^{a(S)-b(S)} (-A^2 - A^{-2})^{|S|-1}$$

State Sum Example: Trefoil

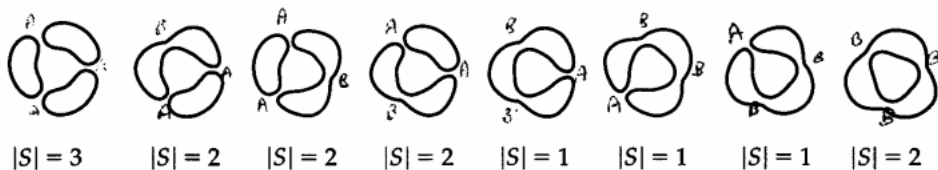
- $n = 3 \Rightarrow 2^3 = 8$ states



$$\langle K \rangle = \sum_S A^{a(S)-b(S)} (-A^2 - A^{-2})^{|S|-1}$$

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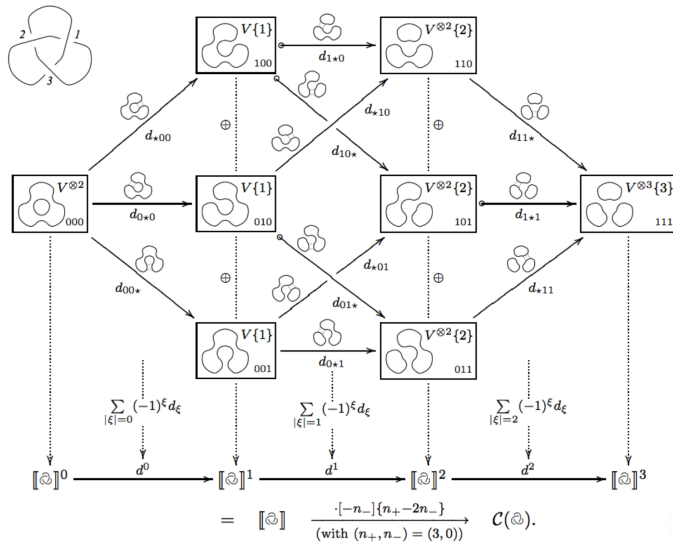
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$$\langle K \rangle = \sum_S A^{a(S)-b(S)} (-A^2 - A^{-2})^{|S|-1}$$

$$\begin{aligned} \langle K \rangle &= A^3 A^0 (-A^2 - A^{-2})^{3-1} + A^2 A^{-1} (-A^2 - A^{-2})^{2-1} + A^2 A^{-1} (-A^2 - A^{-2})^{2-1} \\ &\quad + A^2 A^{-1} (-A^2 - A^{-2})^{2-1} + A^1 A^{-2} (-A^2 - A^{-2})^{1-1} + A^1 A^{-2} (-A^2 - A^{-2})^{1-1} \\ &\quad + A^1 A^{-2} (-A^2 - A^{-2})^{1-1} + A^0 A^{-3} (-A^2 - A^{-2})^{2-1} \\ &= A^3 (-A^2 - A^{-2})^2 + 3A (-A^2 - A^{-2}) + 3A^{-1} + A^{-3} (-A^2 - A^{-2}) \\ &= A^7 - A^3 - A^{-5} \end{aligned}$$

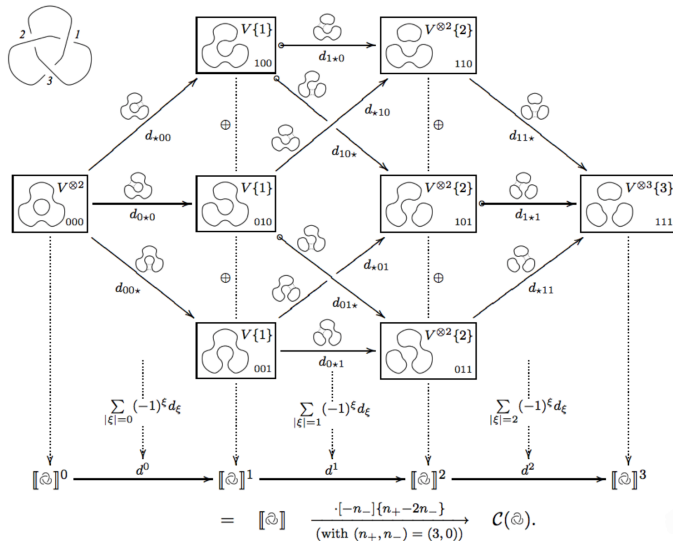
State Hypercube: Trefoil



Why useful

- Organizes 2^n states into a geometric structure
- Replace each state's numerical contribution by a vector space
- Replace local changes by linear maps

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Goal

- Construct a chain complex
- Recover Jones polynomial as Euler characteristic

Cube of Resolutions

- Each resolution is labeled by

$$I = (i_1, \dots, i_n) \in \{0, 1\}^n$$

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- Each edge corresponds to changing a smoothing from 0 to 1
- Hence, moving along an edge increases $r \rightarrow r + 1$.**

From Cube to Algebraic Structure

Goal

Starting from the cube of resolutions, we want to extract meaningful information about the knot.

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Cochain Complex

This structure naturally forms a cochain complex:

$$\dots \xrightarrow{d^{r-2}} C^{r-1} \xrightarrow{d^{r-1}} C^r \xrightarrow{d^r} C^{r+1} \xrightarrow{d^{r+1}} \dots$$

with

$$d^r \circ d^{r-1} = 0.$$

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We extract invariant information via cohomology:

Cohomology

$$H^r = \frac{\ker(d^r)}{\operatorname{im}(d^{r-1})}$$

Example: de Rham Complex on $M \subset \mathbb{R}^3$

de Rham Complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{\nabla} \Omega^1(M) \xrightarrow{\nabla \times} \Omega^2(M) \xrightarrow{\nabla \cdot} \Omega^3(M) \longrightarrow 0$$

(0-forms: scalar functions, 1-forms: vector fields, 2-forms: flux, 3-forms: volume density)

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Cohomology and Euler Characteristic

$$H^k(M) = \frac{\ker(d^k)}{\operatorname{im}(d^{k-1})}, \quad \chi(M) = \sum_k (-1)^k \dim H^k(M)$$

- The Euler characteristic provides a single number that summarizes the topology of the space.

From Circles to ...

Where are we now?

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Key Idea: Replace each circle by a vector space

From Circles to Vector Spaces

Graded Vector Space and q -Dimension

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Assignment to a Circle

$$V = \text{span}\{v_+, v_-\}, \quad \deg(v_+) = 1, \deg(v_-) = -1$$
$$qdim(V) = q + q^{-1}$$

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Multiple Circles

If a state consists of k circles, we assign:

$$V^{\otimes k}$$

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Grading Rule

$$\deg(v \otimes w) = \deg(v) + \deg(w)$$

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Vertex Space

If the resolution I has $k(I)$ circles, define:

$$V(I) = V^{\otimes k(I)}$$

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Vertex Space

If the resolution I has $k(I)$ circles, define:

$$V(I) = V^{\otimes k(I)}$$

- Each vertex now carries a graded vector space

Chain Groups

Height

$$r = |I| = \text{number of 1's}$$

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- Sum over all vertices at the same height
- This forms the r -th level of the complex

Edges: Merge and Split

Local Change Along an Edge

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Idea

The algebraic replacement of these two local moves will be:

$$m : V \otimes V \rightarrow V, \quad \Delta : V \rightarrow V \otimes V.$$

The Maps m and Δ

Multiplication: Merge Map

When two circles merge into one, we use

$$m : V \otimes V \rightarrow V$$

defined by

$$\begin{aligned} m(v_+ \otimes v_+) &= v_+, \\ m(v_+ \otimes v_-) &= v_-, \quad m(v_- \otimes v_+) = v_-, \\ m(v_- \otimes v_-) &= 0. \end{aligned}$$

The Maps m and Δ

Comultiplication: Split Map

When one circle splits into two, we use

$$\Delta : V \rightarrow V \otimes V$$

defined by

$$\Delta(v_+) = v_+ \otimes v_- + v_- \otimes v_+,$$

$$\Delta(v_-) = v_- \otimes v_-.$$

The Maps m and Δ

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$$\Delta(v_+) = v_+ \otimes v_- + v_- \otimes v_+,$$

$$\Delta(v_-) = v_- \otimes v_-.$$

- These are degree -1 maps with respect to the internal grading

Why These Rules? Frobenius Algebra

Key Properties

A Frobenius object consists of maps

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These ensure that merge and split operations are consistent.

From m, Δ to the Differential

Per-edge Map

For each edge ξ , the local change is either a merge or a split. So we define a map

$$d_\xi : V(I) \rightarrow V(I')$$

by inserting either m or Δ in the relevant tensor factors, and the identity on all others.

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Total Differential

The differential is the signed sum of all edge maps:

$$d^r = \sum_{|\xi|=r} (-1)^\xi d_\xi.$$

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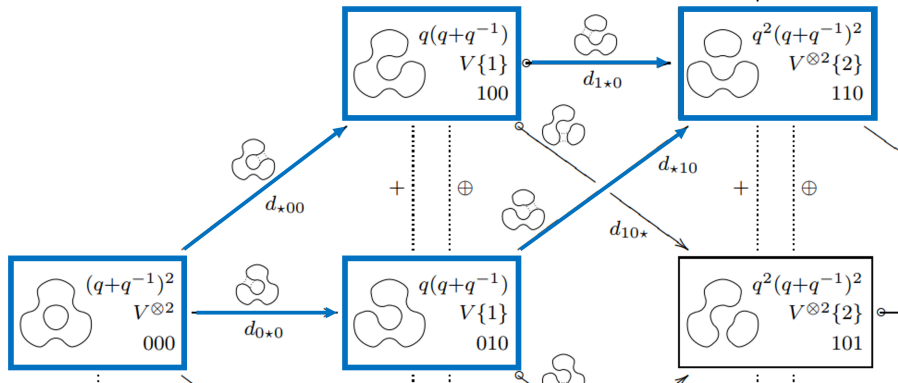
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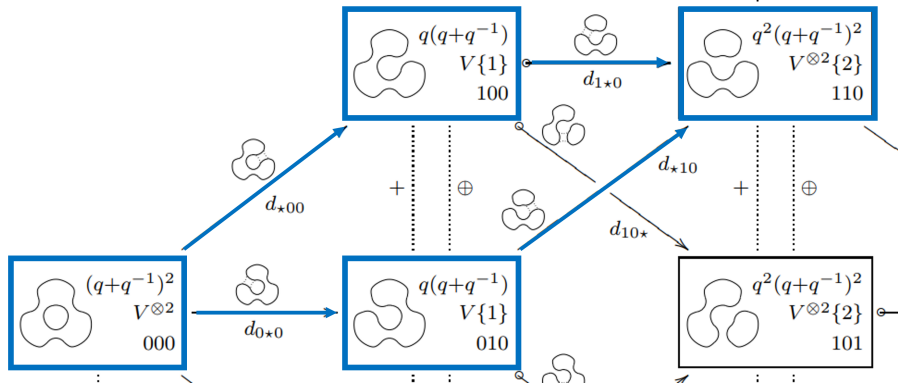
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The Frobenius compatibility and the sign convention ensure $d^{r+1} \circ d^r = 0$.

Differential and Sign Convention



Differential and Sign Convention

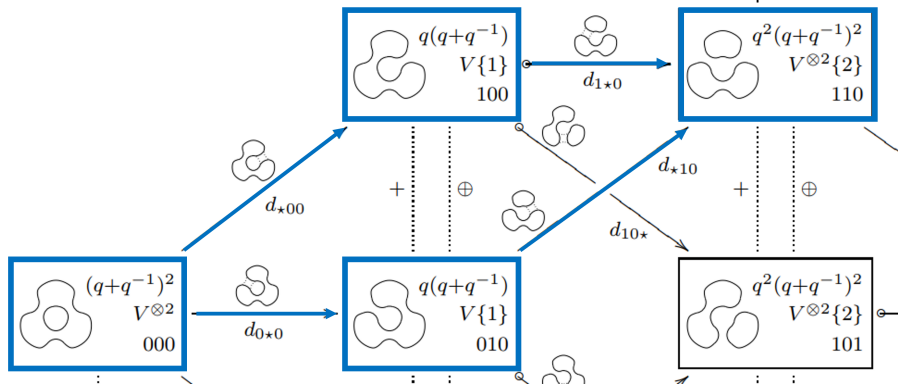


Sign Convention

$$(-1)^{\xi} := (-1)^{\sum_{i < j} \xi_i} \quad (\xi_j = *)$$

(the sign is determined by the number of 1's preceding the position being changed)

Differential and Sign Convention



Key Property

$$d^{r+1} \circ d^r = 0$$

Grading Decomposition and Its Preservation

Grading Decomposition

Each chain group admits a decomposition:

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where j is determined by the degrees of basis elements.

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Examples

(1) $V^{\otimes 2}$:

$$C^{r,2} = \langle v_+ \otimes v_+ \rangle, \quad C^{r,0} = \langle v_+ \otimes v_-, v_- \otimes v_+ \rangle, \quad C^{r,-2} = \langle v_- \otimes v_- \rangle.$$

(2) $V\{1\}$:

$$C^{r,2} = \langle v_+ \rangle, \quad C^{r,0} = \langle v_- \rangle.$$

Degree Shift

Definition

For a graded space V , define:

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Example

If $\deg(v) = 1$, then in $V\{2\}$:

$$\deg(v) = 3$$

Grading Decomposition and Its Preservation

Preservation by d

The differential is built from m and Δ , both of degree -1 .

Each edge also changes the number of circles by ± 1 , introducing a grading shift of $+1$.

$$(-1) + (+1) = 0$$

Grading Decomposition and Its Preservation

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Each edge also changes the number of circles by ± 1 , introducing a grading shift of $+1$.

$$(-1) + (+1) = 0$$

$$\Rightarrow d : C^{r,j} \rightarrow C^{r+1,j}.$$

Summary: Cochain Complex $\mathcal{C}$

From Hypercube to Complex

The **Hypercube** is a collection of states $V(I)$ and edge maps d_ξ . The **Cochain Complex** $\mathcal{C}$ is the algebraic assembly of this data.

- **Nodes to Groups:** We group all spaces $V(I)$ at the same height $|I| = r$ into a single chain group C^r :

$$C^r = \bigoplus_{|I|=r} V(I)$$

- **Edges to Differentials:** We sum all edge maps between height r and $r + 1$ to form a single differential d^r :

$$d^r = \sum_{|\xi|=r} (-1)^\xi d_\xi$$

- **The Complex $\mathcal{C}$:** It is the sequence of these groups and maps:

$$0 \rightarrow C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} C^n \rightarrow 0$$

Summary: Decomposition of Chain Groups: C^r and C_j^r

Internal Grading Decomposition

Each chain group C^r is a **graded vector space**. It can be decomposed into subspaces of fixed internal grading j :

$$C^r = \bigoplus_{j \in \mathbb{Z}} C_j^r$$

- C^r (**The Floor**): The sum of all vertex spaces $V(I)$ at height r .
- C_j^r (**The Room**): The specific subspace of C^r where all basis elements have the **same internal degree j** .
- **The Differential d** : Since $\deg(d) = 0$, it maps C_j^r directly to C_j^{r+1} .

$$d : C_j^r \xrightarrow{\text{Preserves } j} C_j^{r+1}$$

Khovanov Homology

Decomposition of the Complex

Since $\deg(d) = 0$, the complex decomposes into independent sub-complexes for each internal grading j :

$$\mathcal{C} = \bigoplus_j \mathcal{C}_j, \quad d : \mathcal{C}_j^r \rightarrow \mathcal{C}_j^{r+1}$$

- **Bigraded Homology:** Khovanov (co)homology is indexed by two integers:
 - ▶ r : The **homological grading** (height $|I|$ in the cube).
 - ▶ j : The **internal/quantum grading** (preserved by the differential).

$$\mathcal{H}^{r,j}(L) = \frac{\ker(d^r|_{\mathcal{C}_j})}{\operatorname{im}(d^{r-1}|_{\mathcal{C}_j})}$$

- **Structural Invariant:** This bigraded group provides a much finer classification of knots than the Jones polynomial.

Quantum Euler Characteristic and $\hat{J}(L)$

Definition: Quantum Euler Characteristic

For a bigraded complex $\mathcal{C} = \bigoplus_{r,j} \mathcal{C}_j^r$, the **Quantum Euler Characteristic** χ_q is:

$$\chi_q(\mathcal{C}) := \sum_{r,j} (-1)^r q^j \dim \mathcal{H}^{r,j}(L)$$

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- **Recovery of $\hat{J}(L)$:** This quantity is exactly the unnormalized Jones polynomial:

$$\hat{J}(L) = \chi_q(\mathcal{C})$$

- **Standard Jones Polynomial $V(q)$:**

$$V(q) = \frac{\chi_q(\mathcal{C})}{q + q^{-1}}$$

- **Why χ_q ?** While the standard χ collapses all information into a single integer, χ_q uses q^j to "preserve" the internal grading j as a polynomial.

Unit and Counit Maps

Additional Structure

In addition to m and Δ , we use two maps:

$$\iota : \mathbb{Q} \rightarrow V, \quad \epsilon : V \rightarrow \mathbb{Q}$$

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These correspond to birth and death of circles.

Positive Reidemeister I: Local Complex

Local Complex

Let W denote the space attached to the diagram outside the local disk. For a positive kink, the local complex is

$$0 \longrightarrow W \otimes V \xrightarrow{\epsilon} W \longrightarrow 0,$$

where

$$\epsilon(v_+) = 0, \quad \epsilon(v_-) = 1.$$

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Decomposition

Since

$$W \otimes V = (W \otimes \langle v_+ \rangle) \oplus (W \otimes \langle v_- \rangle),$$

the complex splits as

$$(0 \rightarrow W \otimes \langle v_+ \rangle \rightarrow 0) \oplus (0 \rightarrow W \otimes \langle v_- \rangle \xrightarrow{\cong} W \rightarrow 0).$$

Positive Reidemeister I: Local Complex

Decomposition

the complex splits as

$$(0 \rightarrow W \otimes \langle v_+ \rangle \rightarrow 0) \oplus (0 \rightarrow W \otimes \langle v_- \rangle \xrightarrow{\cong} W \rightarrow 0).$$

Conclusion

The second summand is contractible, so only

$$W \otimes \langle v_+ \rangle$$

survives. Since $\deg(v_+) = 1$,

$$W \otimes \langle v_+ \rangle \cong W\{1\}.$$

Hence

$$D_+ \simeq W\{1\}.$$

Negative Reidemeister I: Local Complex

Local Complex

Let W denote the space attached to the diagram outside the local disk. For a negative kink, the local complex is

$$0 \longrightarrow W \xrightarrow{\iota} W \otimes V\{-1\} \longrightarrow 0,$$

where

$$\iota(1) = v_+.$$

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Decomposition

Since

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the complex splits as

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Conclusion

The first summand is contractible, so only

$$(W \otimes \langle v_- \rangle)\{-1\}$$

survives. Since $\deg(v_-) = -1$,

$$(W \otimes \langle v_- \rangle)\{-1\} \cong W\{-2\}.$$

This surviving term sits in cohomological degree 1, so

$$D_- \simeq W[1]\{-2\}.$$

Normalization: Shifting the Complex

Definition of the Normalized Complex $\mathcal{C}(L)$

To make the homology invariant under Reidemeister moves, we apply a **bi-shift** to the raw complex $\underline{\mathcal{C}}(L)$:

$$\mathcal{C}(L) := \underline{\mathcal{C}}(L)[-n_-]\{n_+ - 2n_-\}$$

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- **Homological Shift** $[-n_-]$: Shifts the height r in the hypercube.
- **Quantum Shift** $\{n_+ - 2n_-\}$: Shifts the internal grading j .
- **Effect on q -dimension:**

$$qdim(C^r) = (-1)^{n_-} q^{n_+ - 2n_-} \cdot qdim(\underline{\mathcal{C}}^{r+n_-})$$

- **Purpose:** This "zero-point adjustment" ensures that even if we add twists (crossings) to the diagram, the final homology groups remain unchanged.

Normalization: Shifting the Complex

Why these specific numbers?

These factors are "counter-terms" designed to cancel the grading shifts caused by Reidemeister moves.

- **Positive Twist (n_+):** Adds a factor of $\{1\}$ to the complex. To cancel this, we need a -1 contribution per n_+ .
- **Negative Twist (n_-):** Shifts the homological grading by $[1]$ and the quantum grading by $\{-2\}$.

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- **Negative Twist (n_-):** Shifts the homological grading by $[1]$ and the quantum grading by $\{-2\}$.
- **Total Correction:**
 - ▶ To fix r : We need $[-n_-]$.
 - ▶ To fix j : We combine n_+ and $-2n_- \rightarrow \{n_+ - 2n_-\}$.

"Normalization is the unique gauge-fixing condition that forces the Khovanov complex to be a topological invariant."

Quantum Euler Characteristic

Claim

For a bigraded cochain complex

$$\dots \xrightarrow{d^{r-1}} C^{r,j} \xrightarrow{d^r} C^{r+1,j} \rightarrow \dots$$

with d preserving the j -grading, we have

$$\sum_{r,j} (-1)^r q^j \dim C^{r,j} = \sum_{r,j} (-1)^r q^j \dim H^{r,j}.$$

Quantum Euler Characteristic

Proof

For each (r, j) ,

$$\dim C^{r,j} = \dim \ker(d^r) + \dim \operatorname{im}(d^r).$$

Also,

$$\dim \ker(d^r) = \dim H^{r,j} + \dim \operatorname{im}(d^{r-1}).$$

Hence,

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$$\sum_{r,j} (-1)^r q^j \dim C^{r,j} = \sum_{r,j} (-1)^r q^j \dim H^{r,j} + \sum_{r,j} (-1)^r q^j \dim \operatorname{im}(d^{r-1}) + \sum_{r,j} (-1)^r q^j \dim \operatorname{im}(d^r).$$

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Since shifting $r \mapsto r + 1$ flips the sign,

$$\sum_{r,j} (-1)^r q^j \dim \operatorname{im}(d^r) = - \sum_{r,j} (-1)^r q^j \dim \operatorname{im}(d^{r-1}),$$

From χ_q to the Jones Polynomial

Step 1: From Complex to Polynomial

The quantum Euler characteristic recovers the unnormalized Jones polynomial:

$$\hat{J}(L) = \chi_q(\mathcal{C}) = \sum_{r,j} (-1)^r q^j \dim \mathcal{H}^{r,j}(L)$$

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$$V_L(t) = \frac{\hat{J}(q)}{q + q^{-1}}, \quad q = t^{1/2}$$

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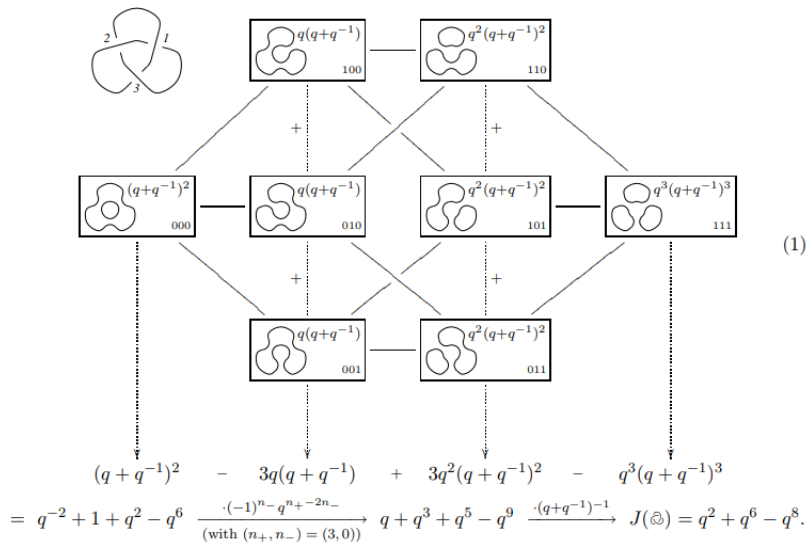
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Example: Trefoil Knot Hypercube



Example: Right-handed Trefoil (3_1)

For a right-handed trefoil, we have:

- **Crossings:** $n_+ = 3, n_- = 0$.
- **Normalization:** $[-n_-] = [0], \{n_+ - 2n_-\} = \{3\}$.

Chain Group q -dimensions (Raw)

- $qdim(C^0) = (q + q^{-1})^2 = q^2 + 2 + q^{-2}$
- $qdim(C^1) = 3(q + q^{-1})\{1\} = 3q^2 + 3$
- $qdim(C^2) = 3(q + q^{-1})^2\{2\} = 3q^4 + 6q^2 + 3$
- $qdim(C^3) = (q + q^{-1})^3\{3\} = q^6 + 3q^4 + 3q^2 + 1$

Calculation of $V(t)$ for Right-handed Trefoil

Step 1: Recovery of $\hat{J}(q)$

From the Khovanov complex, we obtained:

$$\hat{J}(q) = q^3(q^{-2} + 1 + q^2 - q^6) = q^1 + q^3 + q^5 - q^9$$

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Step 3: Final Jones Polynomial $V(t)$

Dividing by the quantum dimension of a circle $(q + q^{-1}) = t^{1/2} + t^{-1/2}$:

$$V(t) = \frac{t^{1/2} + t^{3/2} + t^{5/2} - t^{9/2}}{t^{1/2} + t^{-1/2}} = \mathbf{t} + \mathbf{t}^3 - \mathbf{t}^4$$

Comparison and the Meaning of Span

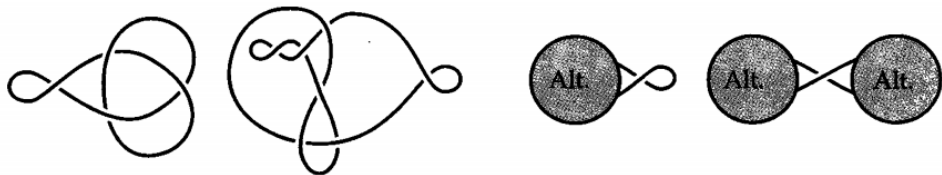
Property	Left-handed (3_1^L)	Right-handed (3_1^R)
Jones $V(t)$	$t^{-1} + t^{-3} - t^{-4}$	$t^1 + t^3 - t^4$
Max / Min Degree	-1 / -4	4 / 1
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- **Span Invariant:** For alternating knots, $\text{Span}(V(t)) = n$. Since trefoil has $n = 3$ crossings, the span is exactly 3.
- **Why q was different?:** In Khovanov homology, q provides a finer "half-step" resolution ($t = q^2$), making the span appear as 6.
- **Chirality:** The polynomials are related by $t \leftrightarrow t^{-1}$, proving 3_1 is a chiral knot.

Reduced Alternating Projection



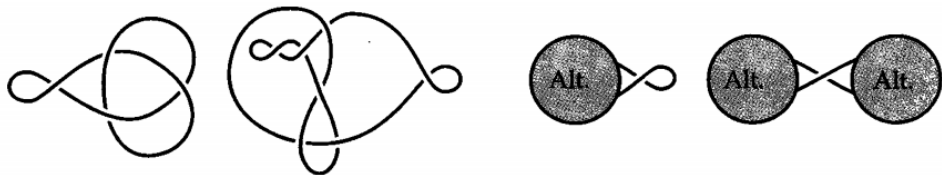
Alternating Diagram

A diagram is **alternating** if, along each component, over- and under-crossings alternate.

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- Alternating + reduced diagrams are “minimal” representations
- These play a central role in Tait’s conjectures

Tait's Conjectures (1898)

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Let D and D' be reduced alternating diagrams of a prime knot K . Then:

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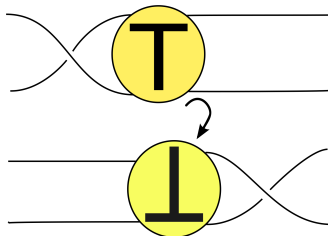
1. **Crossing number is minimized**

$$c(D) = c(D') = c(K)$$

2. **Same writhe**

$$w(D) = w(D')$$

3. **Related by flypes**



Span of Bracket Polynomial

Lemma

If K has a reduced alternating projection with n crossings, then

$$\text{span}(\langle K \rangle) = 4n.$$

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Recall

For any state S ,

$$\langle K \rangle = \sum_S A^{a(S)-b(S)} (-A^2 - A^{-2})^{|S|-1},$$

where

- $a(S)$: number of A-smoothings,
- $b(S)$: number of B-smoothings,
- $|S|$: number of circles in the resolved state.

Proof of Lemma

- For the all-A state S_A :

$$a(S_A) = n, \quad b(S_A) = 0$$

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Key Fact

The span of the bracket polynomial is invariant under Reidemeister moves.

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Tait Conjecture II

All reduced alternating diagrams have the same writhe.

- Follows from the Jones polynomial

Summary and Outlook

Key Achievements

- **Categorification:** Upgraded the combinatorial **Kauffman bracket** to the **Khovanov Homology**, replacing numerical sums with a rich algebraic chain complex.
- **Normalization:** Defined the unique shifts using $n_{\pm}$ to fix the "gauge" and ensure invariance under all Reidemeister moves.
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Furthermore (The "Iceberg")

The Jones polynomial and Khovanov homology point towards:

- **Mathematical Physics:** Statistical mechanics (Yang-Baxter) and Chern-Simons theory.
- **Advanced Algebra:** Braid groups and Quantum groups $U_q(\mathfrak{sl}_2)$.

From numerical invariants to the deep structural data of Khovanov homology.

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Thank you.